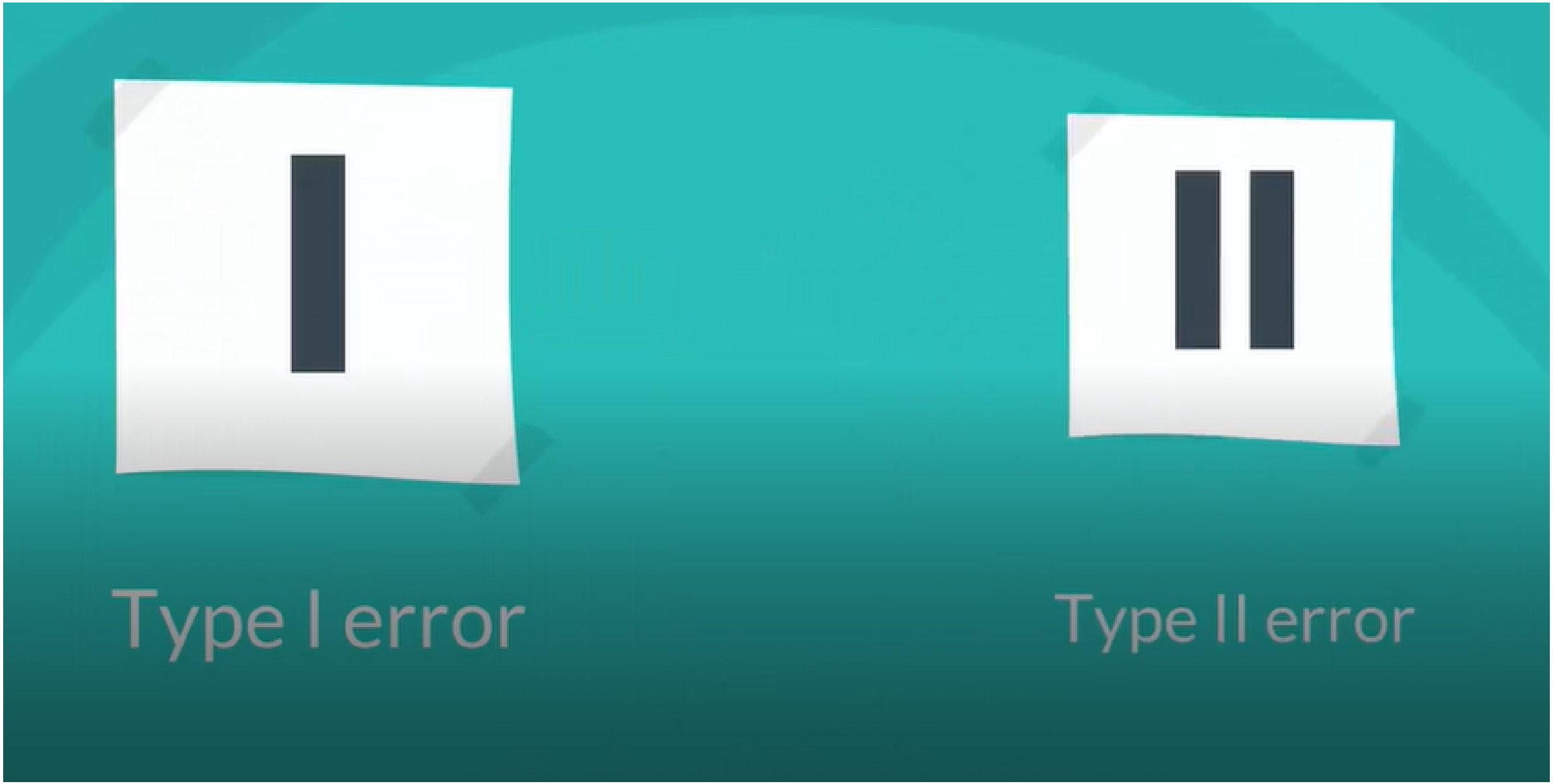
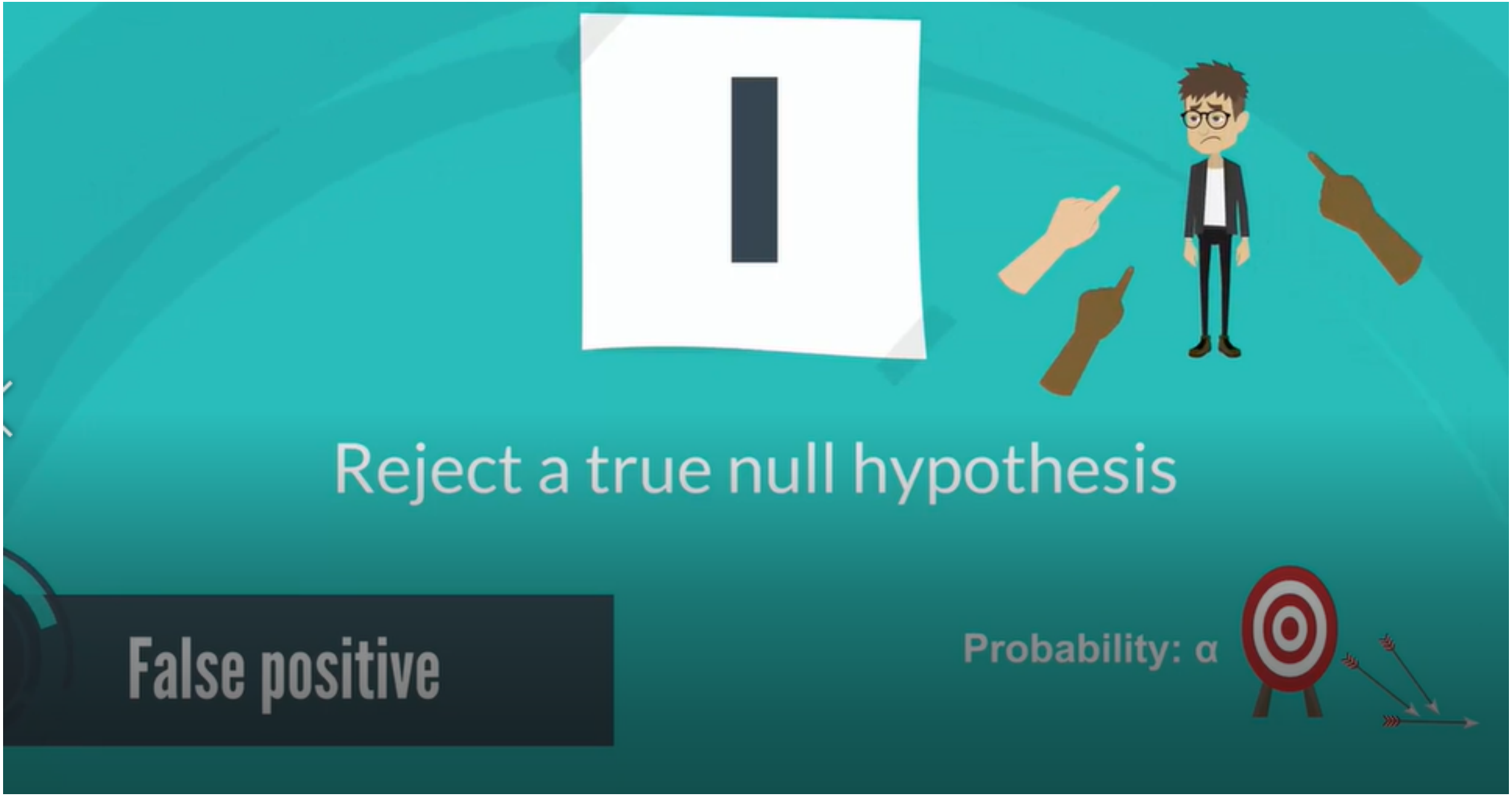
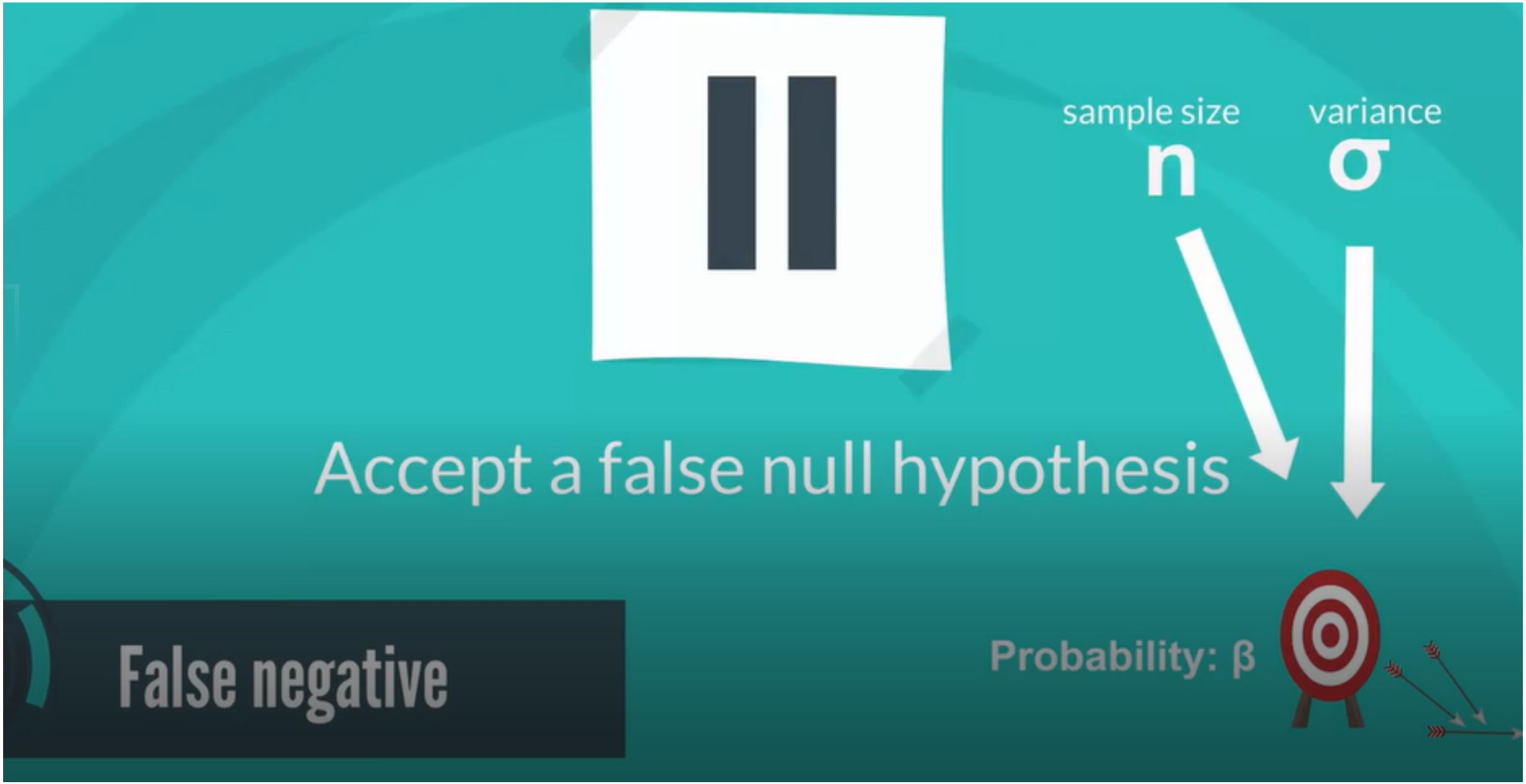

I
II
Type I error
Type II error



Reject a true null hypothesis


I
Reject a true null hypothesis
False positive
Probability: α


II
sample size
n
variance
σ
Accept a false null hypothesis
False negative
Probability: β



# Goal of hypothesis testing

## Rejecting a false null hypothesis

Probability: 1-β



# Goal of hypothesis testing

## Rejecting a false null hypothesis

Probability: 1-β

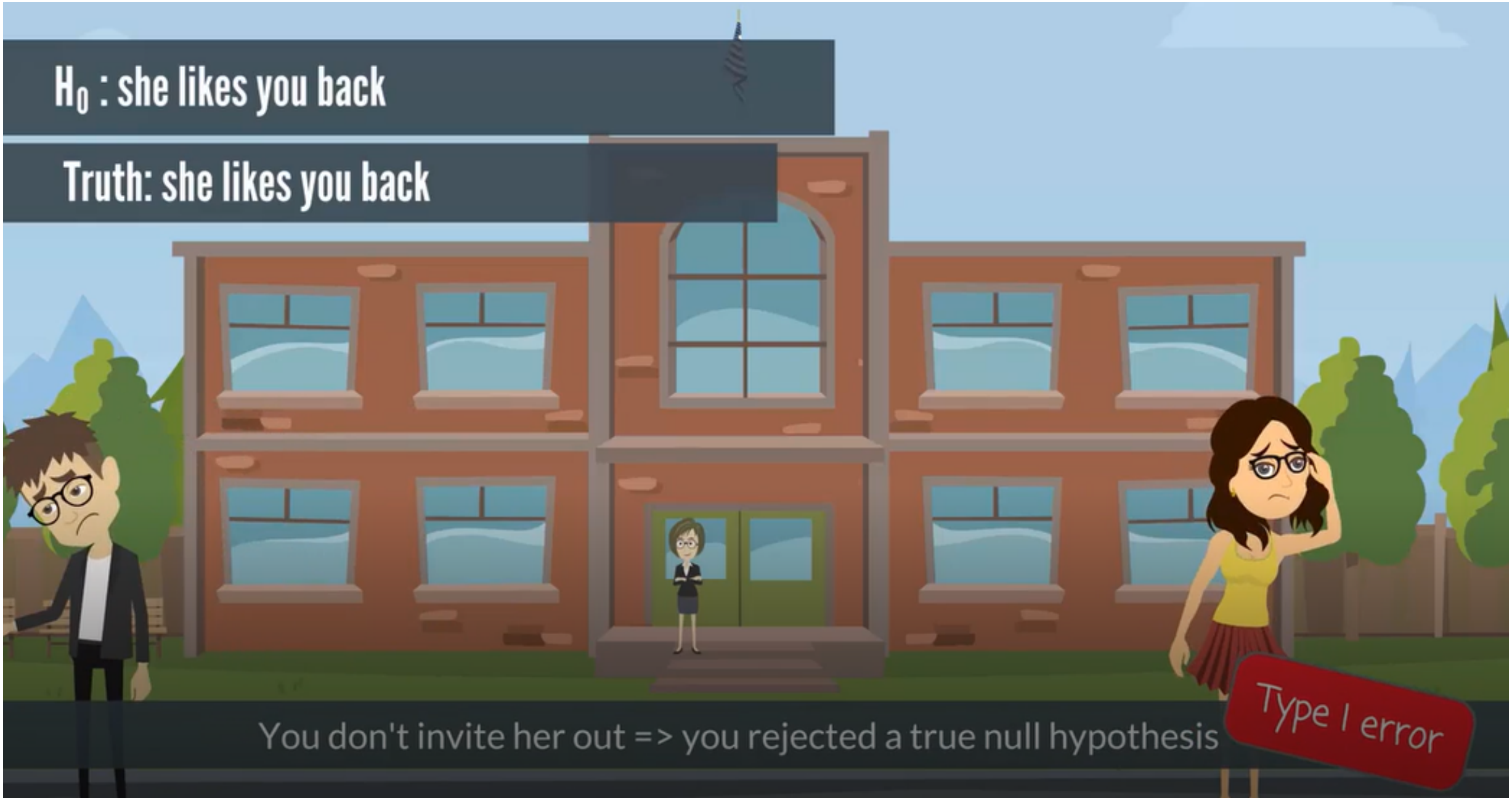
## a.k.a. Power of the test

## Statistical errors

| | | The truth | |
|---|---|---|---|
| | | $H_0$ True | $H_0$ False |
| Your research | Accepted $H_0$ |  | Type II error (False negative) |
| | Rejected $H_0$ | Type I error (False positive) |  |

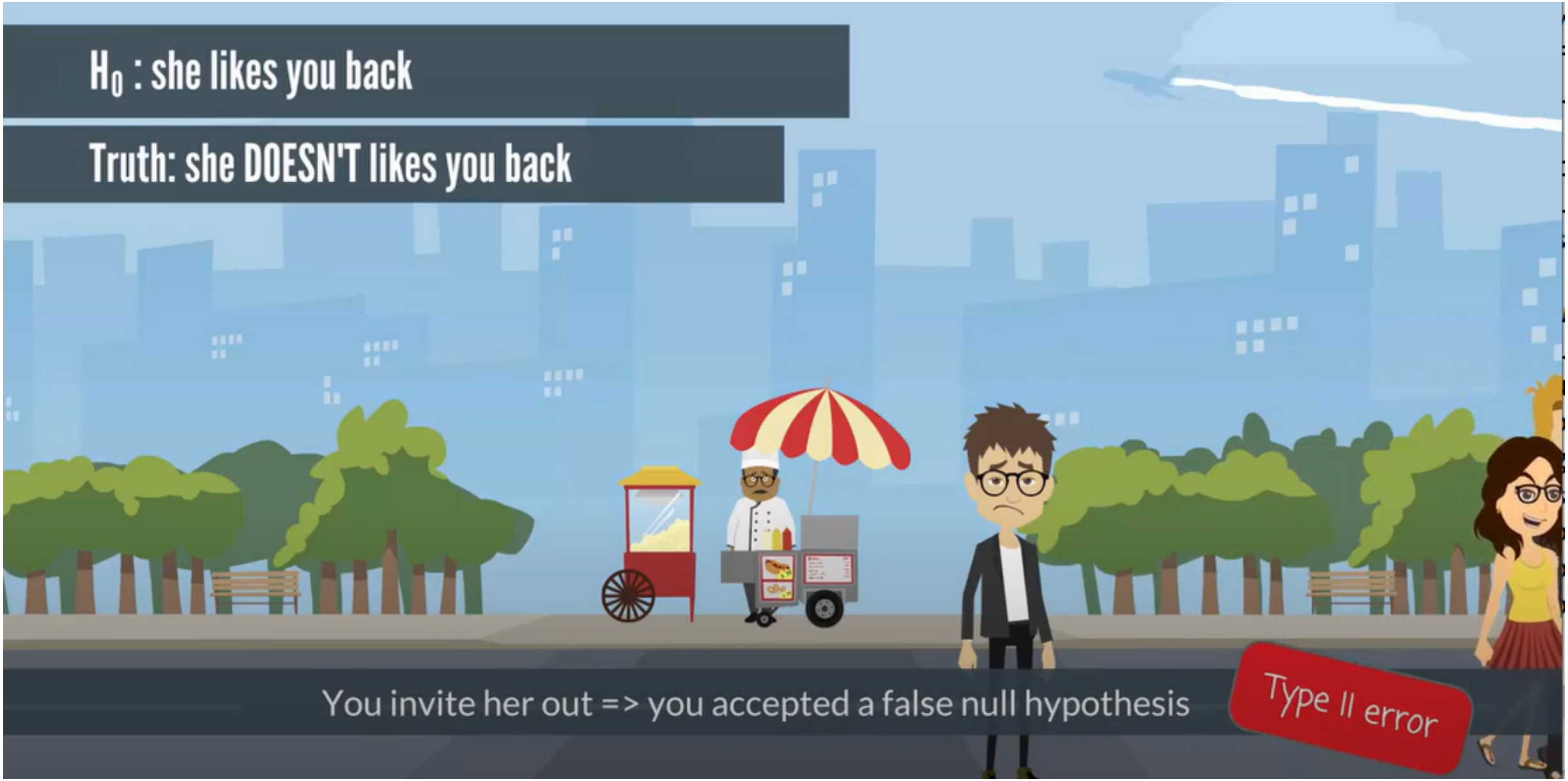




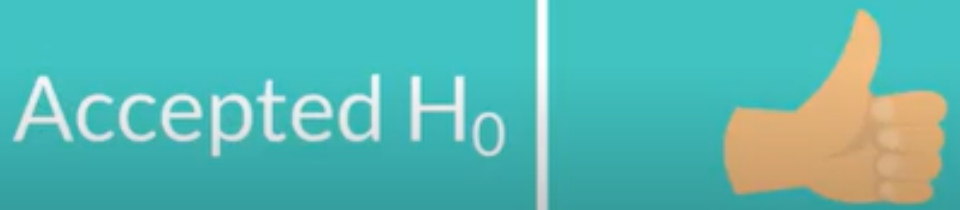

$H_0$ : she likes you back
Truth: she likes you back
You don't invite her out => you rejected a true null hypothesis
Type I error


$H_0$ : she likes you back
Truth: she likes you back
You don't invite her out => you rejected a true null hypothesis
Type I error


$H_0$ : she likes you back
Truth: she DOESN'T likes you back
He's better at statistics than you


$H_0$ : she likes you back
Truth: she DOESN'T likes you back
You invite her out => you accepted a false null hypothesis
Type II error

# Statistical errors

| | | The truth | |
|---|---|---|---|
| | | $H_0$ True | $H_0$ False |
| Your research | Accepted $H_0$ |  | **wrongly invited her**<br>Type II error<br>(False negative) |
|  | Rejected $H_0$ | **missed your chance**<br>Type I error<br>(False positive) |  |